A Survey of Single-Cell Foundation Models for Cell Type Annotation and Perturbation Prediction

InteractiveSurvey

Abstract

Single-cell RNA sequencing has revolutionized the understanding of cellular heterogeneity, driving a paradigm shift from descriptive analysis to predictive modeling through the emergence of foundation models. These large-scale architectures offer the potential to unify diverse datasets and simulate dynamic biological processes, overcoming longstanding challenges related to technical noise and batch effects. This survey critically examines single-cell foundation models specifically designed for precise cell type annotation and accurate perturbation prediction, emphasizing their capacity to forecast counterfactual cellular states. We synthesize recent methodological advances into a cohesive framework that integrates robust statistical foundations, deep generative architectures, and interpretable feature selection strategies. Specifically, we analyze how hierarchical Bayesian deconvolution and simulation-based experimental design ensure stability against distribution shifts, while Variational Autoencoders and Graph Neural Networks facilitate latent space disentanglement and causal inference within gene regulatory networks. Furthermore, we evaluate reinforcement learning approaches for biomarker discovery and argue for specificity-centric evaluation metrics that prioritize mechanistic fidelity over mere reconstruction accuracy. By identifying current limitations in generalization and standardization, this work provides a comprehensive roadmap for developing next-generation models capable of accelerating drug discovery and elucidating complex gene regulatory mechanisms without prohibitive experimental costs.

1 Introduction

Single-cell RNA sequencing has revolutionized our understanding of cellular heterogeneity, enabling the dissection of complex biological systems at unprecedented resolution. As the volume of single-cell data expands exponentially, the field is transitioning from descriptive analyses to predictive modeling, driven by the emergence of foundation models capable of learning universal representations of cellular states. These large-scale architectures promise to unify diverse datasets, overcoming technical noise and batch effects that have long plagued genomic analysis. By capturing the underlying manifold of gene expression, such models provide a robust substrate for simulating biological processes, thereby shifting the paradigm from observing static snapshots to dynamically forecasting cellular behaviors under various conditions.

This survey focuses specifically on single-cell foundation models designed for two critical downstream tasks: precise cell type annotation and accurate perturbation prediction. While early applications primarily addressed clustering and labeling, recent advancements have extended these capabilities to model the causal effects of genetic and chemical interventions. The central thesis of this review is that modern foundation models, when integrated with rigorous statistical frameworks and causal inference mechanisms, can reliably predict counterfactual cellular states. We examine how these systems disentangle basal cell identity from perturbation-specific responses, offering a powerful toolkit for drug discovery and the elucidation of gene regulatory networks without the prohibitive costs of exhaustive wet-lab experimentation.

To establish a rigorous basis for these predictive tasks, we first explore the statistical frameworks essential for handling the inherent sparsity and noise of longitudinal single-cell data. This includes an analysis of robust regression techniques and hierarchical Bayesian deconvolution methods that ensure stability against distribution shifts and technical artifacts. Furthermore, we discuss the critical role of simulation-based experimental de-

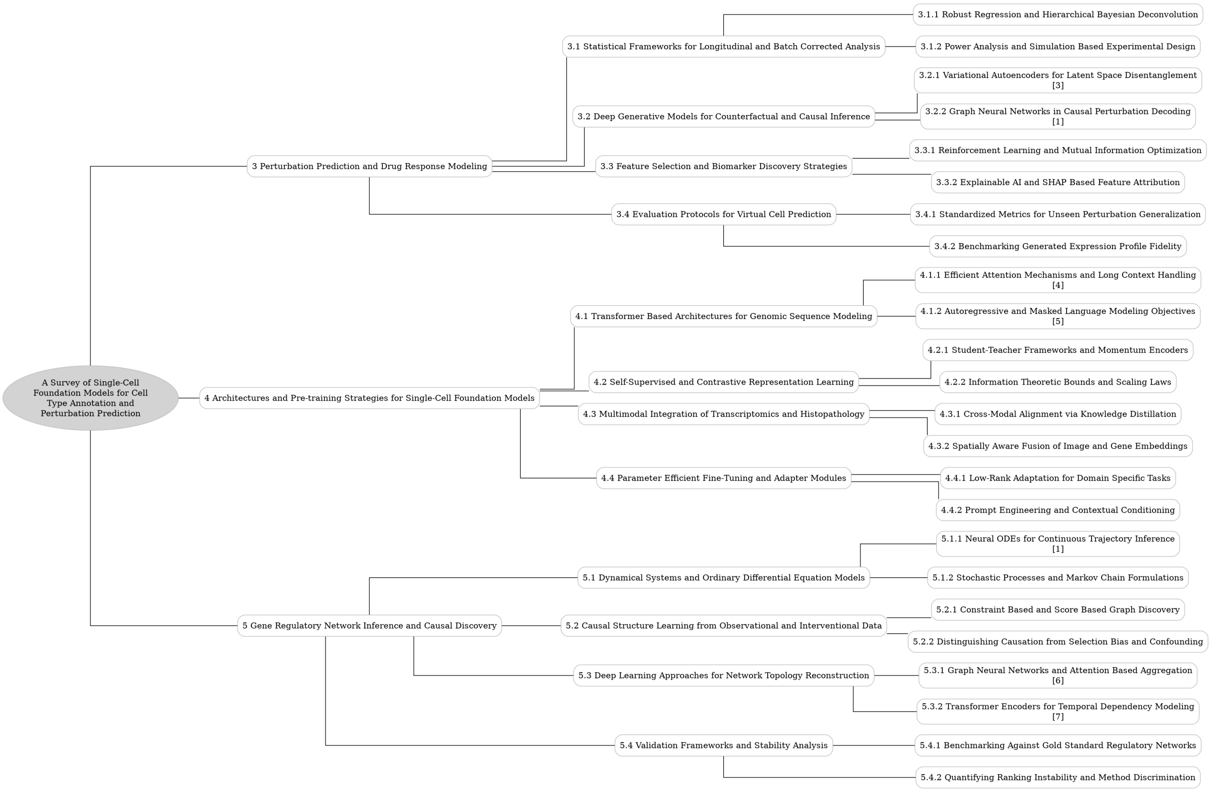

Figure 1: The outline of our survey: A Survey of Single-Cell Foundation Models for Cell Type Annotation and Perturbation Prediction

sign and power analysis in validating model performance. By synthesizing ultra-large datasets with known effect sizes, researchers can benchmark statistical sensitivity and optimize hyperparameters, ensuring that observed improvements reflect genuine architectural advancements rather than data-specific idiosyncrasies.

Building upon these statistical foundations, we delve into deep generative architectures that facilitate counterfactual and causal inference. We provide a detailed examination of Variational Autoencoders engineered for latent space disentanglement, which separate cell identity from perturbation effects to enable precise in silico interventions. Complementing this, we analyze Graph Neural Networks that explicitly incorporate gene regulatory topology to decode causal mechanisms [1]. These models move beyond correlation by propagating signals through biological networks, distinguishing direct drivers from downstream effects, and enhancing the generalizability of predictions to unseen combinatorial interventions and diverse cell types.

Finally, we address strategies for feature selection, biomarker discovery, and the standardization of evaluation protocols. We review approaches

$$c_i^{\text{out},(t+1)} = \frac{1}{|V|-1} \sum_{j \in V} \mathbb{1}\big((i,j) \in E^{(t+1)}\big),$$

Figure 2: Chart from 'Applications of temporal graph learning for predicting the dynamics of biological systems' (arXiv:2605.28659)

leveraging reinforcement learning and mutual information optimization to align latent representations with biological outcomes, alongside Explainable AI techniques like SHAP that attribute predictions to specific genetic drivers. Crucially, we critique current evaluation metrics, arguing for specificity-centric measures that assess the fidelity of predicted regulatory mechanisms rather than mere reconstruction accuracy. This comprehensive overview bridges the gap between black-box predictions and mechanistic understanding, ensuring that computational discoveries are both interpretable and biologically valid.

The primary contribution of this survey is the synthesis of disparate methodological advances into a cohesive framework for single-cell perturbation modeling. We provide a critical assessment of how statistical robustness, causal architecture, and interpretable feature selection converge to enable reliable virtual cell prediction [2]. By identifying current limitations in generalization and evaluation, this work outlines a clear roadmap for future research, guiding the development of next-generation foundation models that can accurately simulate complex biological responses and accelerate therapeutic discovery.

$$\mathrm{LogFC}(p) = \frac{\widehat{\mathrm{lfc}}_p \cdot \mathrm{lfc}_p}{\|\widehat{\mathrm{lfc}}_p\|_2 \; \|\mathrm{lfc}_p\|_2}.$$

Figure 3: Chart from 'Benchmarking virtual cell models for in-the-wild perturbation response' (arXiv:2604.27646)

2 Perturbation Prediction and Drug Response Modeling

2.1 Statistical Frameworks for Longitudinal and Batch Corrected Analysis

2.1.1 Robust Regression and Hierarchical Bayesian Deconvolution

Robust regression techniques in single-cell analysis address the inherent noise and sparsity of gene expression data by minimizing reconstruction errors through specialized loss functions. Methods often employ mean squared error combined with sparsity penalties or zero-inflated negative binomial distributions to effectively model dropout events and technical variation. By enforcing constraints such as unit-normalized decoder weights or utilizing focal loss, these models enhance the stability of gene representation learning, ensuring that predictions remain accurate even when training data is limited or subjected to distribution shifts.

Hierarchical Bayesian deconvolution extends this robustness by framing cell-type proportion estimation as a probabilistic inference problem. These models leverage complex prior distributions, such as Gaussian mixtures or Dirichlet processes, to capture the underlying biological structure while accounting for batch effects and cellular heterogeneity. Through variational inference or Markov Chain Monte Carlo sampling, the framework approximates posterior distributions over latent variables, allowing for the disentanglement of true biological signals from confounding technical factors without requiring explicit ground-truth proportions in unsupervised settings.

The integration of robust regression with hierarchical Bayesian priors creates a powerful paradigm for handling uncertainty in perturbation studies. By jointly optimizing reconstruction fidelity and adherence to biologically plausible latent spaces, these approaches mitigate the impact of outliers and improve generalization to unseen conditions. This synergy enables precise quantification of transcriptional responses and reliable deconvolution of mixed cell populations, ultimately providing a stable foundation for downstream tasks such as drug response prediction and the identification of perturbation-specific gene regulatory mechanisms.

2.1.2 Power Analysis and Simulation Based Experimental Design

Power analysis in single-cell foundation model evaluation necessitates rigorous simulation-based designs to determine adequate sample sizes for detecting differential expression or perturbation responses. Given the high sparsity and noise inherent in single-cell RNA sequencing data, traditional power calculations often fail; thus, researchers increasingly rely on generative models to synthesize ultra-large datasets that mimic realistic biological variance. These simulations typically employ negative binomial or zero-inflated negative binomial distributions parameterized by real-world references, allowing for the systematic injection of known effect sizes and doublet rates to benchmark statistical sensitivity under controlled conditions.

Simulation frameworks facilitate the assessment of model robustness by generating data across varying complexities, such as differing numbers of cell types, trajectory structures, and dropout probabilities. By leveraging principal component analysis-transformed reference datasets and adding uniform or Gaussian noise, these methods create ground-truth scenarios where the validity of downstream clustering, annotation, and causal tracing can be quantitatively measured. This approach enables the evaluation of metrics like Normalized Mutual Information and Adjusted Rand Index across gold and silver standard datasets, ensuring that observed performance gains are not artifacts of specific data idiosyncrasies but reflect genuine architectural improvements.

Furthermore, simulation-based experimental design is critical for optimizing hyperparameters and computational efficiency without incurring the prohibitive costs of training on massive biological repositories. Researchers utilize these synthetic environments to conduct grid searches on regularization weights and adapter configurations, assessing trade-offs between memory usage, wall-clock time, and convergence stability. By establishing reproducible pipelines that disclose compute resources and failed preliminary experiments, the community can standardize evaluation protocols, ensuring that proposed methods possess sufficient statistical power to generalize across diverse preclinical model systems and large-scale perturbation prediction tasks.

2.2 Deep Generative Models for Counterfactual and Causal Inference

2.2.1 Variational Autoencoders for Latent Space Disentanglement

Variational Autoencoders (VAEs) have emerged as a foundational architecture for achieving latent space disentanglement in single-cell genomics, specifically to isolate distinct biological factors such as cell identity and perturbation effects [3]. By partitioning the latent space into orthogonal subspaces, these models optimize separate representations for basal cellular states and external stimuli, which are subsequently recombined during inference to predict counterfactual outcomes. This structural separation allows for the generation of invariant basal states while capturing specific mechanistic shifts induced by genetic or chemical perturbations, thereby facilitating robust causal inference without requiring explicit gene regulatory network topology.

Advanced implementations enhance this disentanglement through specialized architectural constraints and adversarial training objectives. For instance, frameworks like ChemCPA employ separate embeddings for molecules and cell lines coupled with adversarial classifiers to enforce independence between perturbation and context variables. Similarly, the Sparse Additive Mechanism Shift VAE introduces sparsity constraints to model perturbations as additive shifts in the latent space, ensuring that only significant features of data variation are captured. These approaches often utilize Gaussian mixture priors or hierarchical Bayesian frameworks to mitigate technical noise and dropout events, improving the fidelity of the learned representations compared to standard autoencoders.

Despite their efficacy, traditional VAEs face challenges regarding the information bottleneck and polysemanticity, where single latent dimensions may encode multiple overlapping features. To address this, recent methodologies integrate sparse autoencoder principles or conditional generative mechanisms to refine feature interpretability and scalability. By leveraging deep neural networks for representation learning within a probabilistic setting, these refined VAE variants provide a flexible paradigm for modeling nonlinear bidirectional mappings between control and perturbed states. Consequently, they serve as critical components in modern computational pipelines, enabling accurate prediction of cellular responses to complex, unseen perturbations across diverse cell types.

2.2.2 Graph Neural Networks in Causal Perturbation Decoding

Graph Neural Networks (GNNs) have emerged as pivotal architectures for decoding causal perturbation effects by explicitly modeling the topological constraints of gene regulatory networks [1]. Unlike standard fully connected layers that treat genes as independent features, GNNs incorporate biological priors through message-passing mechanisms, allowing the model to propagate perturbation signals along known physical or co-expression edges. This structural induction enables the differentiation between direct causal drivers and downstream statistical correlations, effectively capturing the non-linear propagation of molecular interventions across the cellular interactome.

In the context of causal perturbation decoding, advanced GNN frameworks integrate sparse mechanism shift modeling to isolate specific regulatory logic from dense latent representations. By leveraging counterfactual reasoning, these models simulate in silico knockouts and predict state transitions that diverge from mere reconstruction objectives. The architecture often employs dual-encoder strategies, utilizing graph convolutions to process network-structured data alongside multi-layer perceptrons for feature-only embeddings, thereby facilitating contrastive learning that distinguishes causal edges from spurious co-activations inherent in single-cell data.

Furthermore, recent methodologies combine

GNNs with foundation models to enhance the generalizability of perturbation predictions to unseen combinatorial interventions. By encoding drug structures as molecular graphs and aligning them with cell-state embeddings, these hybrid systems predict dose-response metrics while mitigating overfitting through regularization techniques such as dropout and batch normalization. This end-to-end formulation ensures that the learned latent space reflects true population shifts induced by perturbations rather than preserving instance-specific noise, offering a robust framework for identifying therapeutic targets and understanding complex biological circuitry.

2.3 Feature Selection and Biomarker Discovery Strategies

2.3.1 Reinforcement Learning and Mutual Information Optimization

Reinforcement learning frameworks in this domain often leverage mutual information (MI) maximization to align latent representations with downstream biological states or perturbation outcomes. By treating the encoding of cellular states as a policy, agents are optimized to maximize the MI between the learned latent variables and specific condition labels, ensuring that the compressed representation retains maximal predictive power regarding cell type or drug response. This approach effectively regularizes the feature space, preventing the collapse of distinct biological signals into ambiguous clusters while facilitating robust transfer learning across heterogeneous datasets.

The optimization process typically involves a variational lower bound on the mutual information, which serves as a tractable surrogate for the intractable true MI. During training, the model minimizes a composite loss function that balances reconstruction fidelity with an MI regularization term, often implemented via contrastive learning objectives or adversarial discriminators. This dual objective forces the encoder to disentangle causal factors of variation from technical noise, as the discriminator attempts to distinguish between joint and marginal distributions of the latent codes and target variables, thereby driving the encoder to produce invariant and informative features.

Furthermore, integrating these MI-driven objectives with policy gradient methods allows for dynamic adaptation to sparse reward signals inherent in single-cell perturbation experiments. The agent learns to navigate the high-dimensional gene expression manifold by receiving feedback based on the informational gain of its trajectory towards a desired perturbed state. This synergy between reinforcement learning and information theory enables the discovery of optimal intervention strategies and enhances the interpretability of the learned dynamics, providing a rigorous mathematical foundation for modeling complex cellular differentiation and resistance continua.

2.3.2 Explainable AI and SHAP Based Feature Attribution

Explainable AI (XAI) has become indispensable for validating deep learning models in computational biology, particularly when predicting complex phenotypes like drug response. While post-hoc attribution methods are widely employed to identify gene-level contributors, they often reveal correlations rather than causal mechanisms. To address this, recent frameworks integrate counterfactual perturbations with attribution scoring, allowing researchers to quantify the specific impact of knocking out individual genes on predicted expression profiles. This approach moves beyond simple feature ranking by simulating biological interventions, thereby enhancing the reliability of in silico hypothesis generation.

Among various attribution techniques, SHAP (SHapley Additive exPlanations) based methods offer a theoretically grounded framework for distributing prediction contributions among input features. By leveraging game theory, SHAP values provide consistent and locally accurate explanations, ensuring that the sum of feature attributions equals the difference between the model output and the average prediction. In the context of genomic data, this methodology facilitates the decomposition of high-dimensional gene expression vectors into interpretable components, highlighting specific genetic drivers that influence model decisions while accounting for complex non-linear interactions inherent in biological systems.

However, applying SHAP to large-scale foundation models presents computational challenges due to the exponential cost of exact Shapley value calculation. Consequently, approximate algorithms utilizing sampling or specialized architectures like Sparse Autoencoders are often adopted to make feature attribution feasible for thousands of genes. These adaptations enable the identification of sparse feature circuits and the tracing of information flow through network layers, effectively bridging the gap between black-box predictions and mechanistic biological understanding. Ultimately, refining these attribution strategies is critical for ensuring that AI-driven discoveries align with established biological knowledge and can guide experimental validation.

2.4 Evaluation Protocols for Virtual Cell Prediction

2.4.1 Standardized Metrics for Unseen Perturbation Generalization

The evaluation of model generalization to unseen perturbations necessitates metrics that transcend simple reconstruction accuracy, focusing instead on the fidelity of predicted distributional shifts. The coefficient of determination (R^2) remains a primary standard for quantifying the proportion of variance in dependent variables predictable from independent inputs, particularly in regression-based perturbation forecasting. However, reliance on global error metrics often obscures specific failures in capturing biological nuance; consequently, complementary measures such as Perturbation Direction Correlation (PDCorr) and differentially expressed gene (DEG) overlap are increasingly adopted to assess whether models correctly identify the directionality and magnitude of transcriptional responses rather than merely approximating average expression levels.

A critical challenge in standardized evaluation lies in distinguishing between the detection of a cellular state shift and the specificity of the underlying regulatory mechanism. Recent analyses indicate that while models may achieve high rates in detecting that a perturbation has occurred, they frequently fail to capture the specific transcription factor activities or causal pathways driving the change. This discrepancy highlights the need for specificity-centric metrics that evaluate whether predicted features correspond to true regulatory consequences rather than generic cell-state artifacts. Without such granular assessment, models risk appearing robust while fundamentally misrepresenting the causal structure of the perturbation response, leading to poor transferability across novel cell lines or chemical conditions.

Furthermore, robust benchmarking frameworks must address out-of-distribution generalization by employing strict non-overlap protocols where training and testing perturbations share no chemical or genetic similarity. Metrics in this domain often utilize pseudobulked expression profiles to mitigate single-cell noise, comparing observed versus predicted vectors across held-out conditions to test zero-shot capabilities. The integration of stability metrics, such as seed robustness and reproducibility across batches, ensures that performance gains are not artifacts of stochastic initialization or dataset bias. Ultimately, a comprehensive metric suite must balance global shift accuracy with mechanistic specificity to reliably guide the development of models capable of navigating the vast, unexplored space of biological perturbations.

2.4.2 Benchmarking Generated Expression Profile Fidelity

Benchmarking the fidelity of generated expression profiles necessitates rigorous preprocessing to align heterogeneous model inputs. Standard protocols involve harmonizing gene sets via zero-padding for missing entries and applying counts per million (CPM) normalization followed by log1p transformation to stabilize variance. This step is critical for models like scGPT and scFoundation, which utilize distinct embedding dimensionalities (512 versus 768) and expect specific input gene universes. Ensuring consistent environmental factors during embedding generation allows for a fair comparison of how well these foundation models preserve biological signal amidst technical noise.

Evaluation metrics extend beyond simple absolute correspondence to capture the structural integrity of the transcriptomic manifold. While mean squared error assesses point-wise accuracy, Spearman rank correlation is frequently employed to evaluate the monotonic relationship between predicted and ground-truth gene rankings, which is vital for identifying differentially expressed genes. Furthermore, structural distance metrics, such as the Frobenius norm of centered expression matrices, quantify how well models maintain batch-specific variability and the relative geometry of perturbation effects. These approaches collectively determine whether generated profiles reflect genuine regulatory shifts or merely co-expression artifacts.

Recent advancements emphasize the distinction between correlational patterns and causal regulatory programs in fidelity assessment. Analyses reveal that many models respond to overall expression shifts rather than specific transcription factor activities, resulting in high structural similarity but low regulatory specificity. Consequently, benchmarking frameworks are evolving to include perturbation prediction objectives that distinguish cause from correlation. By validating against synthetic data from gene regulatory networks and high-confidence knockout datasets, researchers can better ascertain if generated profiles accurately mimic the nuanced dynamics of cellular state transitions under genetic intervention.

3 Architectures and Pre-training Strategies for Single-Cell Foundation Models

3.1 Transformer Based Architectures for Genomic Sequence Modeling

3.1.1 Efficient Attention Mechanisms and Long Context Handling

Standard self-attention mechanisms exhibit quadratic computational complexity relative to sequence length, posing significant barriers for modeling high-dimensional genomic data with thousands of genes. To mitigate this, recent architectures employ induced attention strategies, such as the Perceiver framework, which projects high-dimensional inputs onto a fixed set of latent inducing points [4]. This approach reduces complexity from $O(G^2)$ to $O(G \cdot M)$, where M represents the number of inducing points, enabling the processing of extensive gene panels without prohibitive memory costs. Additionally, sparse attention patterns leveraging prior biological knowledge, such as gene connectivity graphs, further restrict computation to relevant subsets of tokens, preserving signal fidelity while discarding redundant interactions.

Handling long context windows is critical for capturing global dependencies in single-cell transcriptomes, where the lack of natural gene ordering necessitates bidirectional context awareness. Models adapted from BERT-style masked language modeling utilize structured attention masks to prevent information leakage between masked targets during pre-training, ensuring robust representation learning across variable sequence lengths. Empirical studies indicate that training on extended sequences, such as 2,048 tokens, significantly enhances model performance on shorter downstream tasks, suggesting that ex-

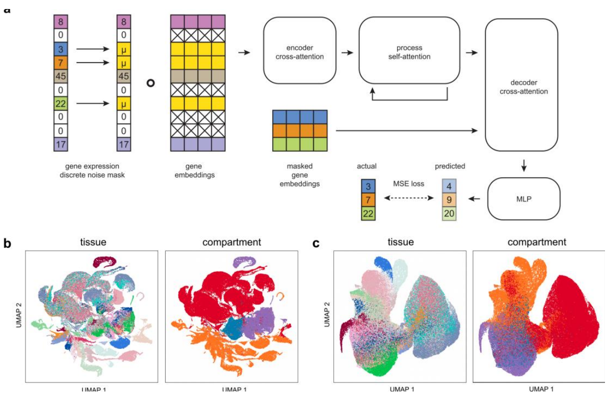

Figure 4: Chart from 'A single-cell gene expression language model' (arXiv:2210.14330)

posure to broader contextual windows improves the understanding of local gene-gene correlations. Techniques like chunk-wise clustering with balanced optimal transport constraints are also employed to maintain uniform cluster sizes, preventing attention dilution in large gene groups.

Advanced optimization techniques further enhance efficiency by addressing input-output awareness and memory bottlenecks. Methods such as FlashAttention utilize tiling strategies to compute exact attention scores with reduced memory footprint and improved IO efficiency, facilitating the training of deeper models on larger datasets. Furthermore, hierarchical modeling approaches replace expensive full self-attention with sparse cross-level message passing, achieving substantial speedups in embedding extraction compared to foundational baselines. These innovations collectively enable the scaling of transformer-based architectures to massive single-cell corpora, allowing for the precise characterization of complex biological states and long-range regulatory interactions that simpler architectures fail to model effectively.

3.1.2 Autoregressive and Masked Language Modeling Objectives

Autoregressive and masked language modeling objectives constitute the foundational pre-training strategies for genomic foundation models, adapting natural language processing paradigms to transcriptomic data. Masked Language Modeling (MLM), inspired by bidirectional architectures like BERT, randomly obscures a subset of gene tokens or expression values, compelling the model to reconstruct them using global contextual information from the entire sequence [5]. This approach is particularly advantageous for single-cell data, which lacks an inherent sequential ordering among genes, allowing the model to capture complex, non-linear dependencies and co-expression patterns without imposing artificial causal constraints.

$$\min_{\pi \in \Pi(\mu,\nu)} \int_{X \times Y} c(x,y)\, d\pi(x,y)$$

Figure 5: Chart from 'SC-OTGM: SINGLE-CELL PERTURBATION MODELING BY SOLVING OPTIMAL MASS TRANSPORT ON THE MANIFOLD OF GAUSSIAN MIXTURES' (arXiv:2405.03726)

In contrast, autoregressive objectives employ causal language modeling to predict subsequent tokens based strictly on preceding context, facilitating iterative generation and sequence synthesis. While standard autoregressive methods assume a linear progression, recent adaptations introduce iterative inference mechanisms where masked elements are predicted sequentially based on model confidence, effectively bridging the gap between bidirectional context utilization and generative capabilities. These hybrid strategies enable models to learn compositional features and robust latent manifolds, enhancing their ability to generalize to unseen cellular states or perturbations through pattern extrapolation rather than mere memorization.

The implementation of these objectives often involves sophisticated loss functions and masking strategies tailored to biological nuances, such as distinguishing between zero-inflation and true absence in expression data. By optimizing for reconstruction accuracy via mean squared error or cross-entropy losses on masked positions, models learn transferable representations that significantly outperform baseline approaches in downstream tasks like cell type annotation and gene regulatory network inference. Furthermore, integrating conditional masking acts as a regularizer, encouraging the learning of invariant condition embeddings that improve performance across diverse datasets and experimental conditions, thereby establishing a scalable framework for multi-omics integration.

3.2 Self-Supervised and Contrastive Representation Learning

3.2.1 Student-Teacher Frameworks and Momentum Encoders

Student-teacher frameworks in representation learning typically employ a dual-network architecture where a student network is optimized via gradient descent while a teacher network provides stable target representations. The teacher encoder does not receive direct gradient updates; instead, its parameters are derived as an exponential moving average (EMA) of the student's weights. This mechanism ensures that the target embeddings evolve slowly over training iterations, effectively decoupling the prediction target from the immediate fluctuations of the student's optimization trajectory and mitigating the risk of representation collapse.

The momentum update rule governs the synchronization between the two encoders, utilizing a momentum coefficient to blend the current student parameters with the historical teacher state. By maintaining a high momentum value, the teacher network acts as a temporally consistent anchor, smoothing the loss landscape for the representation prediction objective. This slowly-evolving target encoder is particularly critical when paired with bidirectional Transformer architectures, as it allows the model to learn robust latent spaces by predicting the teacher's output from perturbed or masked inputs without the instability associated with rapidly changing targets.

Integrating this framework with reconstruction-style losses enables the joint optimization of generative and contrastive objectives within a unified model. The student processes augmented or masked input sequences to match the stable embeddings produced by the teacher from the original data, fostering the learning of invariant features. This architectural design enhances the model's capacity to generalize across diverse biological contexts by ensuring that the learned representations capture essential structural dependencies rather than memorizing transient noise present in individual training batches.

3.2.2 Information Theoretic Bounds and Scaling Laws

Information theoretic bounds provide a rigorous framework for quantifying the intrinsic limits of predictive performance in genomic foundation models. By fitting scaling laws with an additive offset, researchers can estimate the irreducible error floor, which corresponds to the conditional entropy of the data-generating process. This theoretical minimum represents the information content per prediction that no model architecture, regardless of parameter count, can recover due to stochastic biological noise or measurement limitations inherent in single-cell snapshots.

Scaling laws in this domain often deviate from standard power-law behaviors observed in natural language processing, exhibiting metric-dependent regimes where performance may saturate or even degrade as dataset size increases. Phenomena such as retrieval collapse indicate that enlarging the training corpus N can sometimes reduce enrichment scores toward random expectation, suggesting that naive scaling fails to address the long-tailed distribution of rare cell states. Consequently, effective scaling requires balancing model capacity against data complexity to prevent dimensionality collapse and ensure that learned representations capture genuine regulatory structures rather than spurious correlations.

Furthermore, the relationship between model size, computational complexity, and predictive accuracy is modulated by the specific information theoretic metrics employed, such as normalized mutual information or average biological conservation. While increasing parameters generally improves the approximation of the true conditional distribution, the marginal gains diminish rapidly once the model approaches the entropy bound of the target variable. Future architectural developments must therefore prioritize sample efficiency and robustness to sparsity, ensuring that scaling efforts yield meaningful improvements in extrapolation capabilities across combinatorial perturbation spaces rather than merely optimizing for in-distribution reconstruction.

3.3 Multimodal Integration of Transcriptomics and Histopathology

3.3.1 Cross-Modal Alignment via Knowledge Distillation

Cross-modal alignment via knowledge distillation addresses the challenge of integrating heterogeneous biological data by transferring structural knowledge from a teacher network to a student model. Unlike direct contrastive objectives that risk representation collapse, this paradigm employs asymmetric teacher-student architectures where the teacher processes one modality, such as DNA sequences, while the student learns to predict these representations from another, like single-cell transcriptomes. This approach leverages dynamic token adaptation to project embeddings from distinct modalities into a shared latent space, enabling the student to capture complex regulatory logic without requiring paired supervision for every input instance.

The mechanism typically utilizes a masked prediction framework where the student network reconstructs the teacher's latent features from perturbed or masked inputs, effectively shifting the learning signal from explicit embedding matching to latent-space prediction. By minimizing the divergence between the student's output and the teacher's frozen representations, the model enforces semantic consistency across modalities while preserving modality-specific nuances. Structured consistency objectives are often introduced within the encode-flow-decode pipeline to ensure that decoded expressions adhere to biological priors, thereby facilitating smooth transitions between control and perturbed states in the latent manifold.

Recent implementations extend this framework to multi-factor settings, disentangling condition-specific and invariant components to improve generalization across unseen experimental conditions. Methods incorporating prototype-guided regularization further refine this process by aligning cell embeddings with learned class prototypes, enhancing robustness against noise and batch effects. By distilling knowledge from large-scale pretrained foundation models, these techniques enable efficient transfer learning, allowing downstream tasks such as perturbation prediction and batch correction to benefit from rich, context-aware representations derived from diverse data sources without the need for extensive retraining.

3.3.2 Spatially Aware Fusion of Image and Gene Embeddings

Spatially aware fusion architectures address the critical challenge of integrating high-dimensional transcriptomic profiles with spatial context to reconstruct tissue organization. Unlike standard sequence models that treat genes as unordered tokens, these frameworks leverage Joint Embedding Predictive Architectures (JEPA) to predict masked gene expression within a latent space rather than reconstructing raw counts. This approach forces the model to learn dropout-robust features by exploiting the redundancy in cell identity encoding, effectively shifting the learning objective from pixel-level or count-level reconstruction to abstract feature consistency. By modeling gene regulation through coarse-grained representations analogous to image patches, these methods capture both local cellular states and global tissue topology without being hindered by the sparsity inherent in single-cell data.

The core mechanism involves a dual-stream fusion process where gene embeddings and spatial coordinates are mapped into a unified latent manifold. Utilizing adapters or dynamic token adaptation, the model flexibly encodes additional modalities, allowing gene representations to change based on their spatial microenvironment. This is often achieved through attention mechanisms that compute similarity between context vectors, measuring "information-receipt similarity" to identify coordinated gene programs across spatial domains. The integration of structural biological knowledge, such as gene regulatory networks, further refines these embeddings, ensuring that the learned representations preserve biologically coherent modules while capturing remote gene-gene dependencies driven by spatial proximity.

Downstream applications benefit significantly from this spatially enriched embedding space, particularly in zero-shot transfer tasks across diverse tissues and species. By aggregating gene-level embeddings into comprehensive cell-level representations, these models achieve superior performance in cell-type clustering and disease classification compared to methods relying solely on expression counts. The resulting embeddings facilitate the inference of interaction networks and the identification of spatial domains, providing a holistic view of cellular systems. Ultimately, this fusion strategy bridges the gap between high-resolution transcriptomics and structured spatial modules, enabling robust simulation of unseen biological scenarios and enhancing the interpretability of complex multi-omics atlases.

3.4 Parameter Efficient Fine-Tuning and Adapter Modules

3.4.1 Low-Rank Adaptation for Domain Specific Tasks

Low-Rank Adaptation (LoRA) has emerged as a parameter-efficient fine-tuning strategy for adapting large-scale single-cell foundation models to diverse downstream tasks without full parameter retraining. By inserting small, trainable low-rank decomposition matrices into the attention layers of pre-trained transformers, LoRA significantly reduces memory footprint and computational overhead while preserving the integrity of learned biological representations. This approach is particularly advantageous in genomics, where labeled datasets for specific conditions or perturbations are often scarce, enabling rapid adaptation to novel cell types or disease states with minimal data requirements.

In domain-specific applications such as batch correction, cell-type annotation, and perturbation response prediction, LoRA facilitates the integration of biological priors during an intermediate optimization stage prior to task-specific head training. Unlike conventional full-parameter fine-tuning, which risks catastrophic forgetting of generalizable features acquired during pre-training on millions of cells, LoRA restricts updates to a compact set of adapter parameters. This constraint ensures that the model retains robust global transcriptional profiles while flexibly encoding local variations inherent to specific experimental contexts or technological platforms.

Recent implementations demonstrate that LoRA-enhanced models achieve competitive or superior performance compared to task-specific architectures across heterogeneous single-cell RNA sequencing benchmarks. The method supports scalable deployment by allowing a single pre-trained backbone to serve multiple therapeutic and diagnostic applications through distinct, lightweight adapters. Furthermore, combining LoRA with contrastive learning objectives or masked gene modeling during the adaptation phase further enhances representation quality, proving effective in handling long-tailed data distributions and improving generalization to unseen genetic perturbations.

3.4.2 Prompt Engineering and Contextual Conditioning

Prompt engineering in biological foundation models has evolved from simple text templating to sophisticated contextual conditioning mechanisms that integrate heterogeneous data modalities. Unlike standard natural language processing tasks where prompts are purely textual, biological prompting often involves embedding gene expression profiles, protein sequences, or perturbation vectors directly into the input space to guide model inference. This approach leverages the transformer's self-attention architecture to establish long-range dependencies between the conditioning signal and the target prediction, effectively allowing the model to adapt its internal representations to specific cellular states or experimental regimes without extensive parameter updates.

Advanced conditioning strategies frequently employ Classifier-Free Guidance to modulate the influence of supervisory signals during the generative process. By computing a weighted combination of conditional and unconditional predictions, these methods enable precise control over the trade-off between sample diversity and fidelity to the provided biological context. Such techniques are critical for tasks like in-silico perturbation screening, where the model must extrapolate regulatory relationships to unseen drug combinations or genetic modifications. The efficacy of these approaches relies heavily on the quality of the pre-training objective, which must align the representation space with observed measurement distributions while maintaining robustness against technical noise and batch effects.

Despite these advancements, current methodologies face challenges in capturing dynamic, context-specific programs that shift across distinct cell types or disease states. Fixed module granularity and reliance on static priors, such as protein-protein interaction networks, may limit the model's ability to induce novel regulatory structures adaptively. Future directions emphasize the development of learnable, context-adaptive induction mechanisms that allow the program structure to reconfigure based on the input environment. Enhancing the interpretability of attention flows remains paramount, as understanding how specific tokens contribute to contextual understanding is essential for validating biological hypotheses and ensuring the reliability of downstream therapeutic applications.

4 Gene Regulatory Network Inference and Causal Discovery

4.1 Dynamical Systems and Ordinary Differential Equation Models

4.1.1 Neural ODEs for Continuous Trajectory Inference

Neural Ordinary Differential Equations (Neural ODEs) have emerged as a pivotal framework for modeling continuous cellular dynamics, addressing the inherent sparsity and asynchronicity of single-cell transcriptomic data. Unlike discrete-time recurrent architectures, Neural ODEs parameterize the derivative of the hidden state with a neural network, enabling the integration of gene expression trajectories over arbitrary time intervals. This continuous-time formulation allows for the reconstruction of smooth developmental paths from static snapshots, effectively inferring pseudotime-resolved gene regulatory networks without requiring dense temporal sampling or synchronized cell populations [1].

Recent advancements extend linear ODE assumptions to non-linear dynamics by coupling biophysically motivated constraints with deep learning modules. By leveraging differentiable solvers, these models forecast future expression states from initial conditions while simultaneously learning the underlying regulatory mechanics through gradient-based optimization. Techniques such as Virtual Value Perturbation and Gradient Trajectory analysis further refine this process by quantifying instantaneous regulatory influences via counterfactual reasoning, ensuring that inferred interactions remain consistent across varying baseline expression levels and capturing local sensitivity within the regulatory landscape.

Despite their expressive power, training Neural ODEs for large-scale networks presents challenges regarding computational complexity and stability. To mitigate these issues, contemporary approaches incorporate Jacobian regularization and adaptive cross-attention mechanisms that dynamically weight regulatory signals based on node centrality. Furthermore, integrating structure-aware priors from co-expression graphs enhances the model's ability to generalize to unseen time points and robustly infer causal relationships. This synthesis of continuous dynamical systems with graph-based structural knowledge establishes a rigorous foundation for predicting complex cellular differentiation processes and perturbation responses.

4.1.2 Stochastic Processes and Markov Chain Formulations

Stochastic processes provide a rigorous framework for modeling the inherent noise and temporal dynamics of gene regulatory networks, often formalized through Stochastic Differential Equations (SDEs) or Piecewise-Deterministic Markov Processes (PDMPs). In these formulations, molecular concentrations evolve via deterministic drift terms governed by reaction kinetics, coupled with stochastic diffusion components that capture intrinsic biological variability. Specifically, PDMPs model systems where continuous variables, such as protein levels, follow ordinary differential equations that switch randomly based on discrete promoter states, effectively describing the bursty nature of transcriptional activity without assuming continuous differentiability across all state transitions.

Markov chain formulations further discretize cellular state spaces to infer probabilistic transitions along developmental trajectories, treating gene expression snapshots as nodes within a dynamic graph. By constructing transition matrices based on local similarity metrics or random walk operators, these methods estimate the likelihood of cells moving between distinct regulatory states over pseudotime. This approach allows for the derivation of stationary distributions and hitting probabilities, which are critical for identifying stable cell types and transient intermediate states, while simultaneously accounting for the uncertainty inherent in single-cell measurements through probabilistic state assignments rather than deterministic clustering.

Advanced inference algorithms leverage these stochastic foundations to reconstruct time-varying network topologies, extending static Bayesian networks into dynamic regimes capable of capturing feedback loops and non-linear dependencies. Techniques such as variational inference approximate intractable posterior distributions over network parameters by optimizing lower bounds on the marginal likelihood, enabling scalable learning from high-dimensional time-series data. Furthermore, by integrating counterfactual reasoning and gradient trajectory analysis, modern frameworks can disentangle causal regulatory influences from correlational noise, ensuring that the inferred Markovian dynamics accurately reflect the underlying biological mechanisms driving cell fate decisions and response to perturbations.

4.2 Causal Structure Learning from Observational and Interventional Data

4.2.1 Constraint Based and Score Based Graph Discovery

Constraint-based approaches infer graph topology by evaluating conditional independence tests among variables to prune edges that lack statistical support. These methods rely on the assumption that the underlying data distribution faithfully represents the causal structure, utilizing algorithms to identify d-separation sets that orient edges while adhering to Markov properties. Although theoretically rigorous in recovering skeleton structures, their application to high-dimensional genomic data is often hindered by the exponential complexity of testing large conditioning sets and sensitivity to noise in single-cell measurements.

In contrast, score-based methods formulate structure learning as a combinatorial optimization problem, searching the space of possible directed acyclic graphs to maximize a goodness-of-fit metric such as the Bayesian Information Criterion or likelihood scores. These techniques employ heuristic search strategies, including greedy hill-climbing or simulated annealing, to navigate the super-exponential search space efficiently. Recent advancements integrate differentiable relaxation techniques, allowing the discrete structure learning problem to be solved via continuous optimization, thereby scaling effectively to genome-scale regulatory networks where traditional discrete searches fail.

Hybrid frameworks increasingly combine these paradigms to leverage the structural guarantees of constraint-based pruning with the global optimization capabilities of score-based refinement. By initially restricting the search space using conditional independence constraints, these models reduce computational overhead while maintaining robustness against spurious correlations common in omics data. Furthermore, integrating prior biological knowledge as hard constraints or regularization terms within the scoring function enhances the interpretability of inferred gene regulatory networks, ensuring that the discovered topologies align with established mechanistic pathways.

4.2.2 Distinguishing Causation from Selection Bias and Confounding

Distinguishing genuine causal regulatory mechanisms from spurious correlations induced by selection bias and confounding remains a paramount challenge in gene network inference. Observational single-cell data frequently harbors unmeasured latent factors, such as batch effects or cell-cycle stages, which create dependencies that mimic direct regulatory interactions. Standard correlation-based approaches often fail to disentangle these indirect associations, leading to the identification of non-biological links where genes appear connected solely due to shared responses to external stimuli or technical artifacts rather than direct causality.

To address these ambiguities, recent methodologies have shifted toward causal representation learning and interventional frameworks that explicitly model counterfactual scenarios. Techniques leveraging CRISPR-based perturbations provide ground-truth anchors, allowing algorithms to differentiate between co-expression driven by common confounders and actual cause-effect relationships. Furthermore, advanced statistical models now incorporate latent variable estimation to adjust for unwanted variation, utilizing strategies like discrepancy-based variational autoencoders to isolate true regulatory signals from noise inherent in heterogeneous biological samples.

Despite these advances, the risk of false positives persists when perturbation data is sparse or when off-target effects introduce new confounding layers. Robust inference increasingly relies on integrating prior knowledge with data-driven discovery, employing partial correlation analyses and conditional independence tests to prune indirect edges. Future directions emphasize the development of uncertainty-aware models that can quantify the reliability of inferred networks, ensuring that predicted regulatory edges reflect stable biological mechanisms rather than transient statistical artifacts arising from specific experimental conditions or population substructures.

4.3 Deep Learning Approaches for Network Topology Reconstruction

4.3.1 Graph Neural Networks and Attention Based Aggregation

Graph Neural Networks (GNNs) have emerged as a pivotal framework for inferring Gene Regulatory Networks (GRNs) by modeling genes as nodes and regulatory interactions as directed edges [6]. Within this paradigm, message-passing mechanisms aggregate information from neighboring nodes to update latent gene embeddings, effectively capturing both local topology and global structural patterns. While standard convolutional approaches utilize fixed aggregation functions like summation, they often struggle with the inherent sparsity and heterogeneity of single-cell data, necessitating more adaptive strategies to distinguish significant regulatory signals from noise.

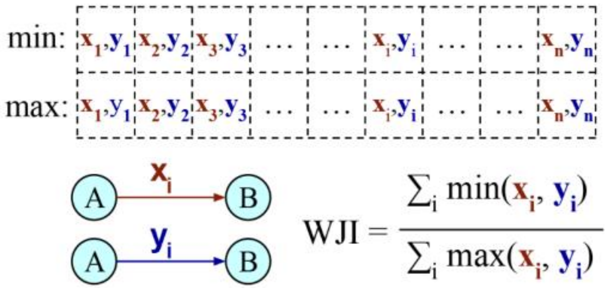

Figure 6: Chart from 'Approaches for benchmarking single-cell gene regulatory network inference methods' (arXiv:2307.08463)

Attention-based aggregation addresses these limitations by introducing learnable coefficients that dynamically weight the contribution of neighboring nodes during message passing. Unlike uniform aggregation, Graph Attention Networks (GATs) allow the model to focus selectively on the most biologically relevant interactions, effectively prioritizing strong transcriptional regulations over weak or spurious correlations. Recent evaluations indicate that architectures employing neighbor sampling, such as GraphSAGE, can further enhance computational efficiency while improving response prediction accuracy compared to isomorphic variants, demonstrating the critical role of specialized aggregation in handling high-dimensional genomic graphs.

Beyond immediate neighborhood aggregation, attention mechanisms facilitate the interpretation of layer-specific biological structures, where early layers often encode protein-protein interactions and deeper layers capture complex transcriptional regulations. However, studies suggest that attention-derived networks do not always yield incremental predictive value over simple statistical baselines for certain tasks, highlighting the need for rigorous validation of attention weights against ground-truth biology. Consequently, current research increasingly explores hybrid approaches that combine the structural inductive biases of GNNs with the flexibility of transformer-based attention to mitigate issues like over-smoothing and to better model long-range dependencies in gene regulatory landscapes.

4.3.2 Transformer Encoders for Temporal Dependency Modeling

Transformer encoders have emerged as pivotal architectures for modeling temporal dependencies in gene regulatory networks by leveraging self-attention mechanisms to capture long-range interactions across time-series transcriptomic data [7]. Unlike traditional recurrent models, these encoders process sequential gene expression profiles in parallel, utilizing positional encodings to preserve temporal order while identifying complex, non-linear regulatory dynamics. This capability allows the model to discern how specific transcription factors influence target genes over varying time lags, effectively mapping the trajectory of cellular state transitions without the vanishing gradient problems inherent in sequential processing.

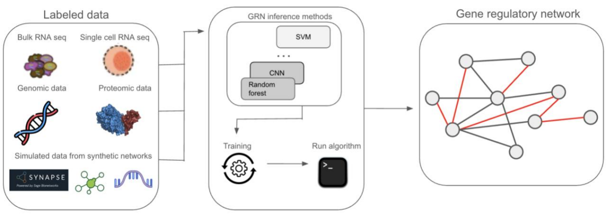

Figure 7: Chart from 'Machine Learning Methods for Gene Regulatory Network Inference' (arXiv:2504.12610)

The internal representation learning within these encoders exhibits a distinct depth-dependent hierarchy, where early layers retain fine-grained relational details of specific regulatory events, and deeper layers distill this information into robust, categorical distinctions of gene programs. By extracting hidden states from intermediate transformer blocks, researchers can isolate features corresponding to different stages of biological processes, such as the emergence of trajectory-relevant markers during differentiation. This stratified processing enables the model to simultaneously handle immediate regulatory responses and cumulative temporal effects, providing a multi-scale view of dynamical systems that static models fail to capture.

Furthermore, adaptation strategies for temporal tasks often involve modifying standard pretraining objectives to incorporate explicit temporal constraints or utilizing attention weights as proxies for regulatory strength between time points. Approaches include fusing structural embeddings with encoder outputs to guide decoding processes or employing masked language modeling objectives tailored to predict future expression states based on historical context. These methods facilitate the construction of dynamical models that not only infer network topology but also simulate system behavior under perturbations, offering a powerful framework for predicting cell fate and understanding the temporal logic underlying genetic regulation.

4.4 Validation Frameworks and Stability Analysis

4.4.1 Benchmarking Against Gold Standard Regulatory Networks

Benchmarking gene regulatory network (GRN) inference against gold-standard networks remains the definitive method for assessing algorithmic accuracy and generalizability. This process typically involves comparing predicted adjacency matrices against curated ground-truth interactions derived from extensive perturbation studies, such as those found in the Adamson, Norman, and Dixit datasets. Evaluation metrics predominantly focus on the Area Under the Precision-Recall Curve (AUPRC) and Pearson correlation coefficients of differential expression, which quantify the model's ability to distinguish true regulatory edges from non-interactions while capturing magnitude changes in post-perturbation states.

Recent comparative analyses reveal significant disparities in performance when models face heterogeneous data distributions or sparse sampling conditions. Traditional tree-based and regression methods often struggle to generalize beyond their training contexts, frequently yielding AUPRC scores near random guessing when applied to unseen cellular states. In contrast, emerging foundation models and geometric approaches demonstrate superior robustness, leveraging large-scale pretraining to extract intrinsic regulatory mechanisms that remain invariant across datasets. These advanced architectures effectively map intergene knowledge to ground-truth networks, achieving substantial performance gains even in challenging settings involving specific transcription factors or metabolic regulators.

The rigorous validation against gold standards also highlights the critical importance of data quality control and meta-cell aggregation in ensuring reliable benchmarking outcomes. By integrating high-quality transcriptomic profiles with known regulatory topologies, researchers can systematically evaluate context-dependent representations and the stability of inferred edges under varying activation states. Such standardized pipelines not only facilitate direct comparison of alternative strategies but also drive the development of universal GRN inference paradigms capable of translating single-cell foundation model embeddings into accurate, biologically meaningful regulatory graphs for therapeutic discovery.

4.4.2 Quantifying Ranking Instability and Method Discrimination

Quantifying ranking instability is essential for distinguishing robust biological signal from stochastic noise in gene regulatory network inference. Traditional aggregate metrics like AUROC often mask local volatility, particularly when positive samples are sparse or data distributions shift. To address this, recent frameworks employ bootstrap resampling and label permutation to assess the variance of edge confidence scores across iterative runs. By normalizing rank differences between regulatory and non-regulatory pairs, researchers can derive stability indices that reveal whether a method consistently prioritizes true interactions or merely fits dataset-specific artifacts.

Method discrimination relies heavily on rank-based metrics that evaluate the specificity of predicted perturbations rather than binary classification accuracy. Metrics such as Hit@1% and early precision ratios focus on the top-tier predictions, which are critical for downstream experimental validation where resources limit testing to a few candidates. These approaches effectively penalize models that suffer from mode collapsing or indiscriminate batch mixing, phenomena that can yield deceptively high global scores while failing to capture meaningful biological hierarchies. Consequently, rank-specific evaluation provides a more rigorous benchmark for comparing algorithms under imbalanced regulatory scenarios.

Furthermore, the decoupling of ranking performance from latent space dimensionality challenges the assumption that low-rank representations inherently improve inference. Empirical analyses indicate that correlations between effective rank and predictive performance are often confounded by layer depth, suggesting that architectural choices influence ranking stability independently of compression rates. By integrating stability bounds with gene-level influence scores, modern evaluation protocols can audit the generalizability of new methods. This multifaceted approach ensures that reported improvements reflect genuine discriminative power rather than overfitting to specific data splits or noise structures.

5 Future Directions

Despite significant advancements in single-cell foundation models, several critical limitations persist that hinder their widespread adoption in clinical and discovery settings. Current architectures often struggle with generalization to unseen cell types or combinatorial perturbations, frequently relying on training data distributions that fail to capture the full spectrum of biological diversity. While variational autoencoders and graph neural networks offer promising mechanisms for disentanglement and causal inference, they remain susceptible to the information bottleneck and polysemanticity, where latent dimensions encode overlapping features rather than distinct biological factors. Furthermore, existing evaluation protocols predominantly prioritize reconstruction accuracy over mechanistic fidelity, lacking standardized, specificity-centric metrics that rigorously assess whether predicted regulatory mechanisms align with ground-truth biology. The reliance on static snapshots also limits the ability to model dynamic temporal trajectories, leaving a gap in understanding the evolution of cellular states post-perturbation.

Future research must prioritize the development of dynamic, time-aware foundation models that explicitly incorporate temporal dependencies to simulate the continuum of cellular responses rather than isolated endpoints. Integrating ordinary differential equations or continuous-time normalizing flows within graph neural network frameworks could enable the precise modeling of gene regulatory dynamics following intervention. Additionally, there is an urgent need to enhance model interpretability by coupling sparse mechanism shift modeling with advanced explainable AI techniques, ensuring that predictions can be traced back to specific causal drivers within the gene regulatory network. The community should also establish unified benchmarking suites based on large-scale, multi-modal synthetic datasets generated through rigorous power analysis. These benchmarks must evaluate models not only on label prediction but also on their ability to recover known causal edges and predict dose-response curves for novel drug combinations. Finally, incorporating active learning loops where computational predictions guide iterative wet-lab validation will create a feedback mechanism to continuously refine model priors and reduce uncertainty in low-data regimes.

Addressing these challenges will fundamentally transform the landscape of therapeutic discovery and systems biology. By shifting from descriptive correlation to reliable causal prediction, next-generation models will drastically reduce the cost and time associated with preclinical drug screening, allowing researchers to virtually test thousands of combinatorial interventions before entering the laboratory. Enhanced interpretability will bridge the gap between black-box algorithms and mechanistic understanding, fostering trust among biologists and clinicians while accelerating the identification of novel biomarkers and therapeutic targets. Ultimately, the realization of robust, dynamic virtual cell models will enable a paradigm shift towards precision medicine, where treatments are tailored based on accurate in silico forecasts of individual patient cellular responses, thereby optimizing efficacy and minimizing adverse effects.

6 Conclusion

This survey has synthesized the rapidly evolving landscape of single-cell foundation models, emphasizing their transition from descriptive clustering tools to predictive engines capable of simulating counterfactual cellular states. We examined how robust statistical frameworks, including hierarchical Bayesian deconvolution and simulation-based experimental design, provide the necessary rigor to handle the inherent sparsity and noise of longitudinal data. Furthermore, our analysis highlighted the critical role of deep generative architectures, specifically Variational Autoencoders and Graph Neural Networks, in disentangling basal cell identity from perturbation-specific effects. By integrating causal inference mechanisms with biological priors, these models move beyond mere correlation to decode the underlying regulatory logic driving gene expression changes. Finally, we explored advanced feature selection strategies leveraging reinforcement learning and mutual information optimization, which ensure that latent representations remain biologically interpretable and aligned with specific downstream outcomes such as drug response and biomarker discovery.

The significance of this review lies in its cohesive integration of disparate methodological advances into a unified framework for virtual cell prediction. By bridging the gap between black-box deep learning and mechanistic biological understanding, this work establishes a new standard for evaluating model fidelity based on causal accuracy rather than simple reconstruction metrics. We argue that the convergence of statistical robustness, causal architecture, and interpretable feature selection is essential for overcoming current limitations in generalization across diverse cell types and unseen combinatorial interventions. This synthesis not only clarifies the theoretical underpinnings of modern foundation models but also provides a practical roadmap for researchers aiming to deploy these systems in high-stakes applications like therapeutic target identification and personalized medicine, where reliability and interpretability are paramount.

Looking forward, the field must prioritize the development of standardized evaluation protocols that rigorously test model performance against gold-standard biological benchmarks rather than synthetic proxies alone. Future research should focus on enhancing the scalability of these architectures to accommodate multi-modal data integration, thereby capturing the full complexity of cellular dynamics beyond transcriptomics. We call upon the community to foster greater collaboration between computational scientists and experimental biologists to validate in silico predictions through targeted wet-lab experiments, ensuring a continuous feedback loop that refines model accuracy. By addressing current challenges in data heterogeneity and causal validation, next-generation foundation models will ultimately transform our ability to forecast complex biological responses, accelerating the pace of discovery and reducing the reliance on costly, exhaustive empirical screening.

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